

Industry Insights

Steering Toward the Future: The Use of Piezo Technology for Optical Intersatellite Communication

Yuri Toegemann^{ID}, Piezosystem Jena, Inc., Hudson, MA 01749 USA

INTRODUCTION

Recent years have seen a reemergence of interest and investment in low Earth orbit (LEO) satellite constellations. First proposed in the 1990s, the construction of such constellations was once the hopeful plan of companies, such as Iridium, Globalstar, Odyssey, and Teledesic [1]. Unfortunately, most of these plans never fully came to fruition as the upfront construction costs were unfeasible, and a new competing network implementing fiber optic cables was introduced. These fiber optic cable networks promised users high data rates at a fraction of the construction costs of a satellite network. Today, as worldwide data throughput and coverage requirements increase beyond the capacity of fiber optic cable networks, companies, such as SpaceX, Amazon, ViaSat, OneWeb, and Telesat, are resurrecting the idea of LEO constellations [2]. In order to avoid the fate of their predecessors, these companies must build robust constellations with reliable satellites as launching satellites comes at high costs. One of the most critical systems aboard these satellites is the communication terminal that allows each satellite to communicate with other satellites in the constellation.

As opposed to the old ways of radio-frequency intersatellite communication, many of these new companies propose using optical, specifically infrared, frequencies for data transmission. The increased frequency and bandwidth of the optical signal allow satellites to achieve higher data throughput and the close proximity of satellites in the constellations reduces latency [3]. This level of performance is easier to achieve using the optical spectrum, as using radio frequency communication would require extremely large and power-consuming antennas [4]. However, this

improved system of communication comes with its own challenges. The main obstacle to optical intersatellite communication is the extremely precise beam stabilization and pointing necessary to acquire and maintain laser communication between moving satellites. To overcome this challenge, new optical communication terminals on LEO satellites must rely on tip-tilt mirror mechanisms, called fast steering mirrors, to steer and stabilize these infrared beams.

Piezosystem jena is a global leader in the development of piezo-based flex hinge stages and actuators. Now, piezosystem jena is bringing expertise in tip-tilt stages to partnering satellite developers to build and adapt fast steering mirrors for space use in these next-generation optical intersatellite communication terminals.

WHY USE OPTICAL COMMUNICATION?

DATA SPEEDS AND THROUGHPUT

Satellites communicate by using modulated signals to transmit data in binary code. Today, there are advanced modulation techniques that favor the use of higher frequency signals and larger bandwidths. For this reason, LEO satellite constellation developers plan to use unregulated optical, specifically infrared, frequencies rather than radio and microwave frequency signals historically used for intersatellite communication [5]. Using this wider spectrum of higher frequencies directly allows for more data throughput [6].

DATA SECURITY

Increased data throughput is not the only reason why LEO satellite constellation builders are switching to infrared frequency intersatellite communication. Another benefit of utilizing this range of the electromagnetic spectrum is the ability to avoid interference between signals. While the frequency of infrared signals is much greater than that of radio signals, the beamwidth of infrared signals is much smaller [5]. Thinner beams mean there is a much

Authors' current address: Y. Toegemann, Piezosystem Jena, Inc., Hudson, MA 01749 USA (e-mail: ytoegemann@piezोजना.com).

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smaller chance of two different signals accidentally crossing and interfering with each other. In addition, thinner beams make it much harder for third parties to intercept these signals. This represents an improvement in both reliability and security in transmissions.

ROLE OF FAST STEERING MIRRORS FOR OPTICAL INTERSATELLITE COMMUNICATION

Next-generation satellite communication inherently requires next-generation communication hardware. For LEO satellite constellations, this means new communication terminals capable of transmitting and receiving infrared laser signals are required. A diagram of one such terminal can be seen in Figure 1.

In this example, two fast steering mirrors serve the roles of a fine steering mirror and a point-ahead mirror. The fine steering mirror is the backbone of the operation and is responsible for beam stabilization. Beam stabilization is important because a satellite in LEO experiences disturbances due to atmospheric conditions. In addition, the satellite can produce its own disturbances and vibrations as various mechanical systems operate onboard. The fine steering mirror is responsible for correcting beam misalignment from both sources of disturbances. In order

to perform these corrections, the fine steering mirror must be partnered with at least one sensor, which will provide feedback on the incoming beam's direction. In this example, the sensor used is a 4-quadrant detector [7].

As opposed to the fine steering mirror, the point-ahead mirror completes an equally important and challenging task. The purpose of this fast-steering mirror is to deflect the transmitted beam ahead of the receiving terminal's location in order to account for the time it takes for the beam to travel, and the relative motion between the two satellites. In order to determine the necessary point-ahead angle, the calculation takes place on the transmitting satellite using GPS information for the location of both satellites. This calculated angle can be extremely small, and thus a fast-steering mirror being used for a point-ahead mirror must be extremely accurate [7].

Piezo tip-tilt mirrors are well equipped to serve as both fine steering and point-ahead mirrors. For both roles, the fast-steering mirror is only required to achieve small displacement angles, but at high frequencies. Piezo tip-tilt systems are fully capable of achieving the displacement and frequencies required. For example, the PSH 25 OEM tip-tilt actuator from piezosystem jena can achieve ± 16 mrad in a closed loop and has a resonant frequency of 1400 Hz. Based on the requirements piezosystem jena has received from satellite customers, these specs far surpass

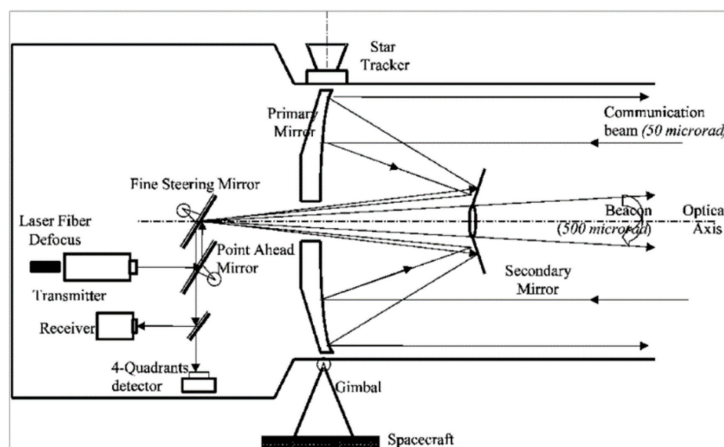


Figure 1.
Diagram of an OISL communication terminal [7].

the speed and displacement requirements needed for use in optical satellite communication terminals. Competing technologies, such as voice coil tip-tilt actuators and MEMS mirrors, provide similar performance but have limitations in other areas. For example, MEMS mirrors are simply too small for use in many satellites. Mirrors used for steering these beams in space must adhere to strict quality requirements that increase their size, as well as being large enough to handle the diameter of the beams. Generally, this results in mirrors needing to be larger than 12-mm diameter, while MEMS mirrors are usually no larger than 7 mm.

Voice coil actuators draw too much power for smaller satellites. Unlike piezo actuators, voice coils use what is known as the Lorentz force to actuate, which results in a proportional relationship between the displacement of the actuator and current draw [8]. Since this current must be constant, the voice coil requires a high power draw. Piezo-electric actuators on the other hand are economical with their power draw as maintaining positions requires only enough current to compensate current leakage, and dynamic operation at small displacements requires minimal power draw as well. Minimizing the power draw of the mirror system is critical in satellites where available power is highly limited, making piezo actuators ideal mirror tip-tilt platforms for use on satellites.

SPACE QUALIFICATION OF FAST-STEERING MIRRORS

Besides fulfilling the deflection and speed requirements for steering the communication beams, fast-steering mirrors for use in satellites must also fulfill space qualification requirements. While there is no universal standard for what it means to be “space qualified,” NASA and ESA both have documents specifying space qualification tests and protocols for space vehicles under their jurisdiction. These include the NASA-STD-7002B [9], GSFC-STD-7000B [10], and the ECSS-E-ST-10-03C [11] documents.

From these three documents, one can deduce four main categories of requirements and tests for space qualification. These categories are material testing, vibration testing, shock testing, and thermal testing. These tests address the main environmental stresses on systems during launch and in operation in space. Piezo tip-tilt mirrors are well equipped to pass these tests and satisfy the requirements.

Material testing: One of the primary concerns of material testing is to determine a component’s susceptibility to outgassing. Outgassing is a process where materials in space, under extreme vacuum conditions, lose mass due to evaporation [12]. For a tip-tilt mirror, outgassed materials can recondense and land on the mirror and the rest of the optical system. This poses a major threat to the durability of the system as it can cause a decrease in the performance

of the laser communication terminal. Therefore, a tip-tilt mirror mechanism for use in space must comprise materials that exhibit very low outgassing.

In order to satisfy this requirement, piezo tip-tilt actuators can easily be modified for use in vacuum and UHV. The modification includes removing rubber insulation and replacing other high outgassing materials with low outgassing materials (ex. adhesives). In addition, manufacturers can bake out piezo elements up to 150 °C to remove volatile compounds that otherwise would have been released in a vacuum.

Another type of material testing is one for radiation tolerance. The most common test is for total ionizing dose (TID), where a component is subjected to the full amount of ionizing radiation expected to accumulate over the mission lifetime [13]. A component passes the TID test and can be expected to fulfill its requirements in space if the component remains functional following this radiation dose. Typical total ionizing doses range between 5 and 10 krad for LEO satellites on a seven-year mission [13]. Piezo tip-tilt mirrors can easily be made to deal with this dose of radiation.

Vibration testing: Vibration testing is necessary in order to ensure the structural integrity of a satellite component during and after launch. There are two main types of vibration testing: 1) random vibration testing and 2) sine sweep vibration testing [13]. Both involve subjecting the component to specified accelerations at different frequencies. Typically, random vibration testing takes place over a range of 20–2000 Hz, while sine sweep testing only takes place over 5–100 Hz [10] and [14].

Piezo actuators from piezosystem jena excel in withstanding these vibrations due to the inherently high stiffness of piezo ceramics, as well as their use of solid-state flex hinges. These hinges are designed using simulation software to indicate internal stresses, allowing engineers to develop strong and robust designs that fulfill strength requirements.

Shock testing: Shock testing typically addresses three kinds of shock that a space component could encounter in space. This includes pyro shock, self-induced shock, and externally induced shock [9]. However, given these three types of shock, only the most extreme needs to be tested.

Piezo actuators must be carefully constructed to deal with shocks. Piezo ceramics are sensitive to tensile and lateral forces, meaning bending forces can cause them to break. A shock to the casing of a normal piezo stack can cause permanent failure. However, if designed in a manner that isolates the piezo stacks from bending forces, piezo tip-tilt actuators can resist failure due to shock. To date, piezosystem jena has tested proprietary tip-tilt actuators at shocks consistent with space qualification, and the results showed no failure or breaking of the piezoceramics or actuator structure.

Thermal testing: Thermal testing for space components is a straightforward process. Based on recent NASA free-

space communication technology solicitations, a tip-tilt system on an LEO satellite would need to operate in temperatures between -20°C and 70°C [15]. In order to test a component for these temperatures, it will be thermally cycled between the minimum and maximum temperatures for at least 100 cycles [13]. As a safety measure, the actual testing temperature range will be $(T_{\min} - 10)^{\circ}\text{C}$ to $(T_{\max} + 10)^{\circ}\text{C}$, meaning in this case a testing range of -30°C to 80°C would be used [13]. This ensures that the component will only experience temperatures in space that were tested and qualified. In addition to thermal cycling in ambient conditions, space components will also have to be thermally cycled in vacuum conditions [11]. During thermal vacuum testing, the component is cycled in the same range of temperatures as the ambient tests only this time at 1×10^{-6} Torr. Again 100 cycles are completed, with failure expected within the first 20 cycles for parts not suitable for this environment [13].

Piezo actuators are easily capable of passing thermal testing as they can operate from -269°C to 150°C . However, on the low end of this range below -20°C , the actuator displacement will decrease as the temperature approaches the minimum operable temperature. Luckily, piezo ceramics are capable of full displacement operation between -20°C and 80°C , which encompasses the -20°C to 70°C range indicated by NASA. This is true for both ambient and vacuum operations.

FUTURE FOR PIEZOSYSTEM JENA

As evidenced by the contents of this publication, piezosystem jena is aware of the many performance and testing requirements for tip-tilt systems used in optical intersatellite communication. piezosystem jena is currently developing space-qualified mirror tip-tilt actuators and has completed a wide range of tests. As such, piezosystem jena looks forward to new partnerships in the design of new tip-tilt systems for use in space.

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